

# ESP32 Driver Board User Manual

(V1.0)

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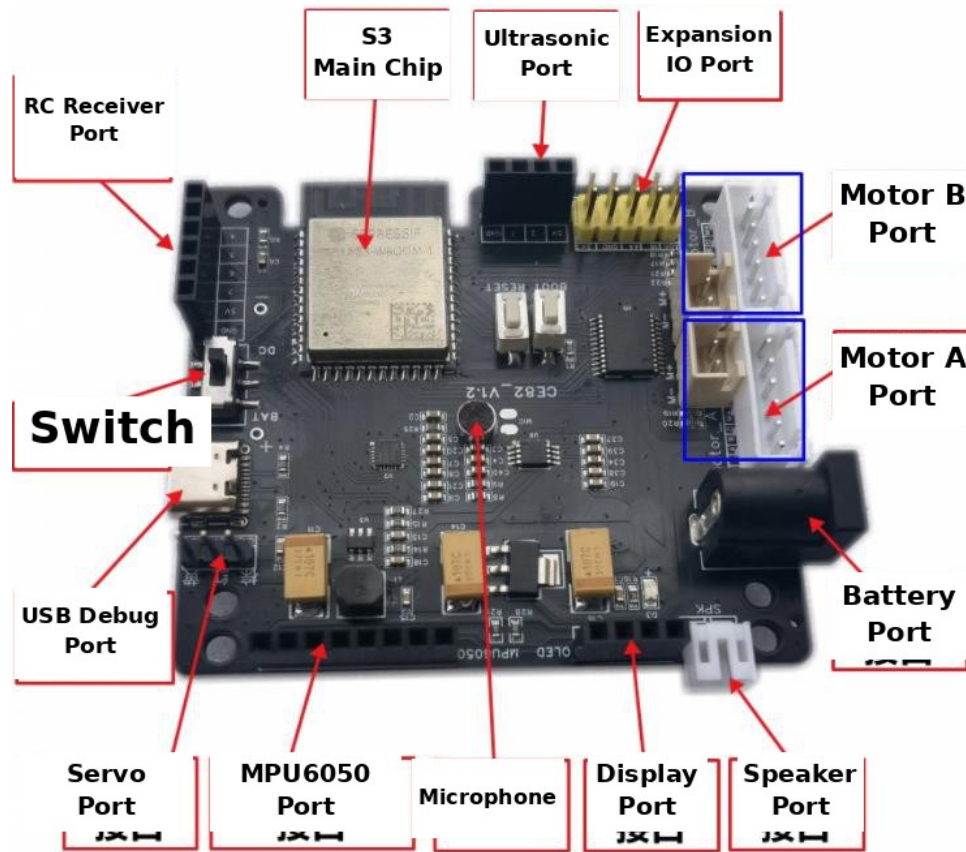
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## 1. Hardware Interfaces

This section introduces the driver board's hardware interfaces. First-time users are required to read it carefully.

An overview of the driver board is shown below:



### 1.1. ESP32 Hardware Resource Allocation

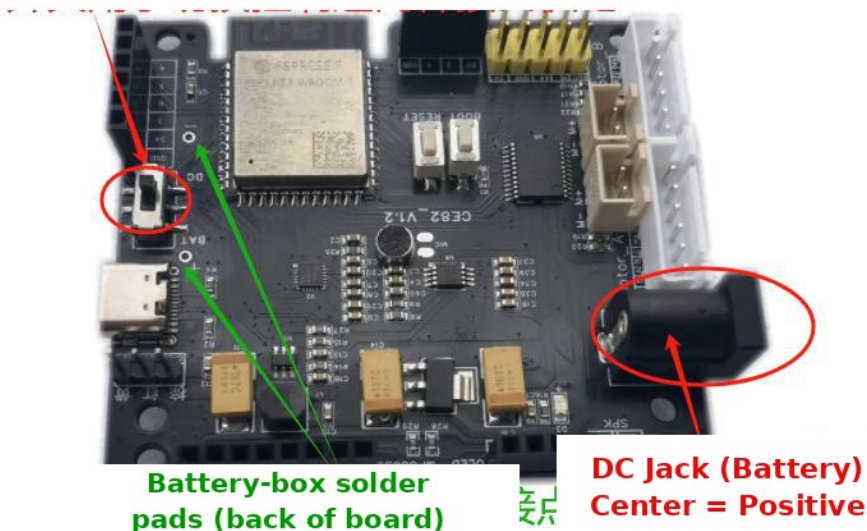
| Module                | Function | ESP32 IO Pin |
|-----------------------|----------|--------------|
| Motor A               | A_IN1    | 45           |
|                       | A_IN2    | 48           |
|                       | A_PWM    | 47           |
|                       | A_ENC_1  | 14           |
|                       | A_ENC_2  | 21           |
|                       |          |              |
| Motor B               | B_IN1    | 38           |
|                       | B_IN2    | 39           |
|                       | B_PWM    | 40           |
|                       | B_ENC_1  | 41           |
|                       | B_ENC_2  | 42           |
|                       |          |              |
| Line-Following Module | S1       | 7            |
|                       | S2       | 6            |
|                       | S3       | 5            |
|                       | S4       | 4            |
| Servo                 | Ch.1     | 15           |
| Display               | SCL      | 13           |

|                         |         |    |
|-------------------------|---------|----|
|                         | SDA     | 12 |
| Download Port (via USB) | DP      | 20 |
|                         | DN      | 19 |
| LED                     | LED1    | 46 |
| Button                  | KEY1    | 0  |
| Battery Level Sense     | ADC     | 3  |
| Ultrasonic              | ECHO    | 1  |
|                         | TRIG    | 2  |
| MPU6050                 | SCL     | 13 |
|                         | SDA     | 12 |
|                         | INT     | 17 |
| Voice Module ES8311     | I2S_SCL | 13 |
|                         | I2S_SDA | 12 |
|                         | I2S_MCK | 8  |
|                         | I2S_BCK | 9  |
|                         | I2S_DI  | 10 |
|                         | I2S_WS  | 11 |
|                         | I2S_DO  | 16 |

## 1.2. Power Supply

The driver board requires a 12V lithium battery for power. The battery can connect via the DC jack or the battery-box solder pads on the back of the main board. Note: the DC jack is center-positive.

**Switch: selects DC-jack or battery-box power**



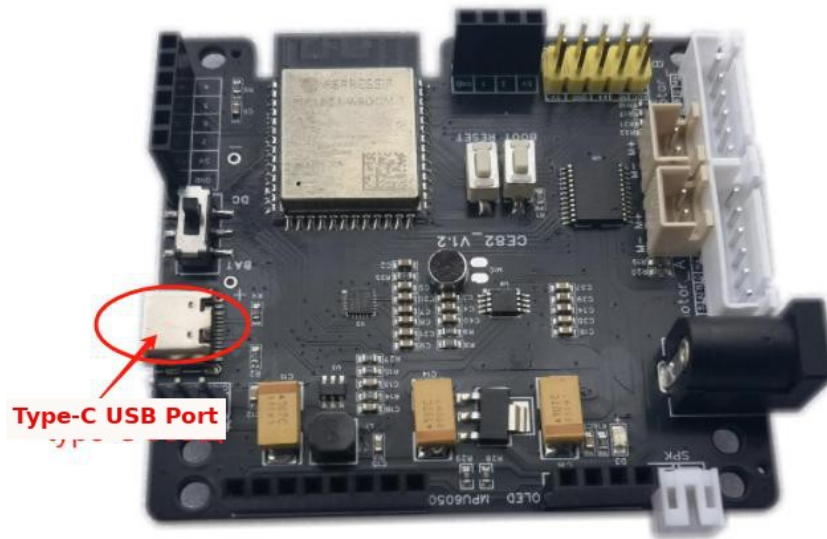
## 1.3. USB Port

The USB port serves two functions: uploading firmware and serial debugging.

When the USB cable is plugged in, the power indicator LED lights up immediately, but the USB port only supplies 5V, so only the 5V-and-below circuits operate normally (the ESP32 system, servo, display, RC receiver port, and ultrasonic port).

The motor ports on the board run on 12V, so a lithium battery is required for them to work properly — running the motors from USB power alone will make them spin much more slowly.

Serial debugging means using the S3's USB port to output LOG data.



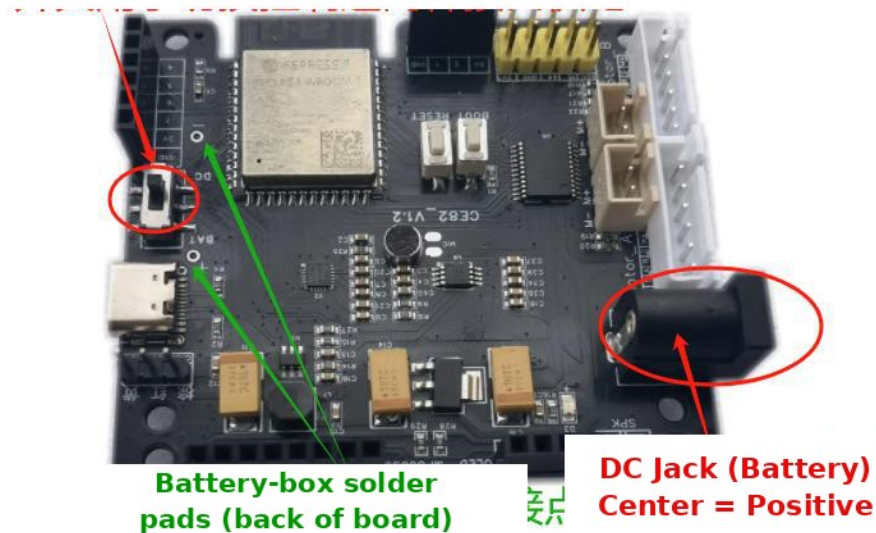
## 1.4. Switch

The switch toggles between battery-box power and DC-jack power.

Switch set to the BAT side: powered from the battery box.

Switch set to the DC side: powered from the external DC jack.

**Switch: selects DC-jack or battery-box power**

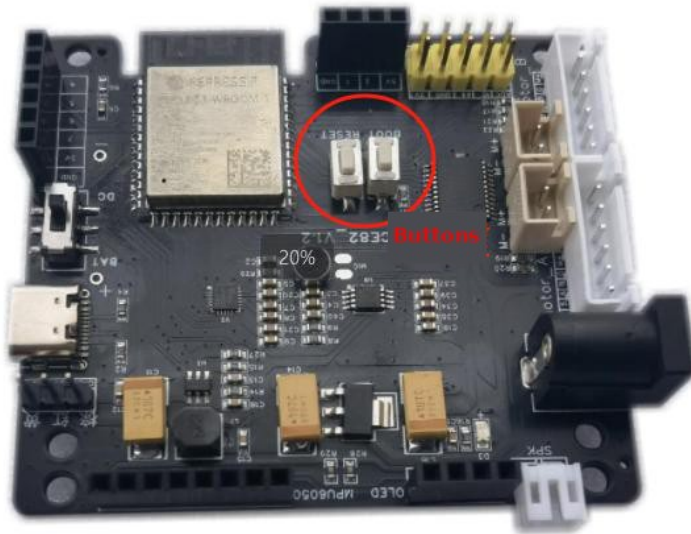


## 1.5. Buttons

The dev board has 2 buttons: a Reset button and a BOOT button.

The Reset button resets the ESP32 system.

The BOOT button is user-programmable; holding it at power-on enters download (flashing) mode.



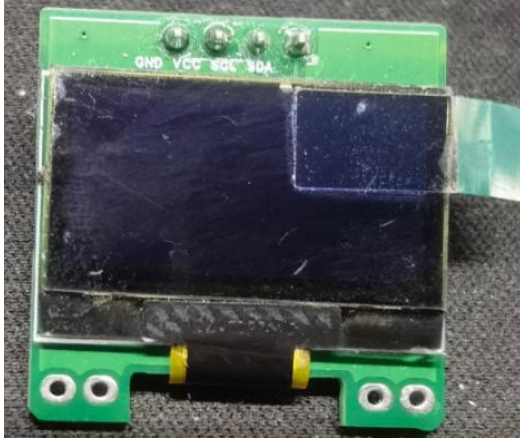
## 1.6. LED

The dev board provides one power indicator LED. It lights up to show the board is powered.

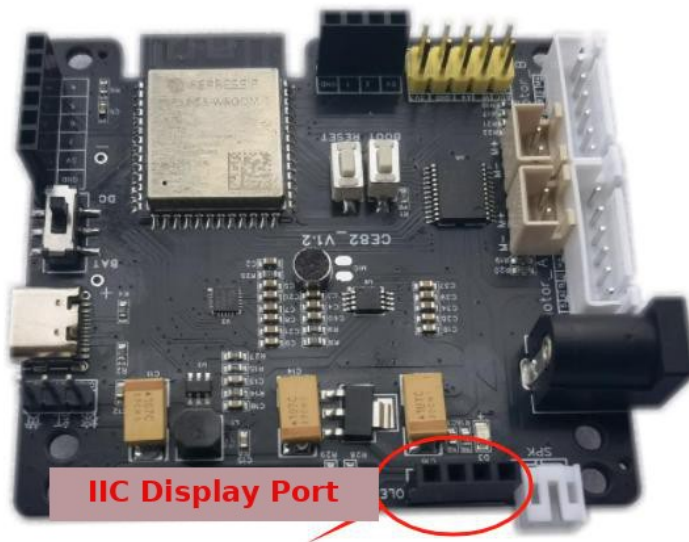


## 1.7. Display Port

Supports a 4-wire IIC display, as shown:

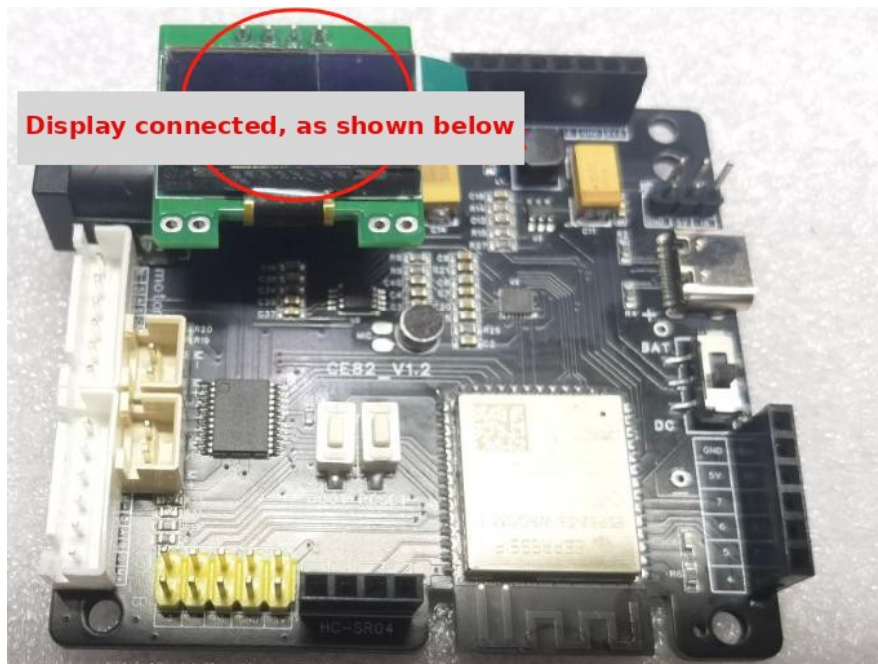


The LCD port on the ESP32 dev board is shown below:



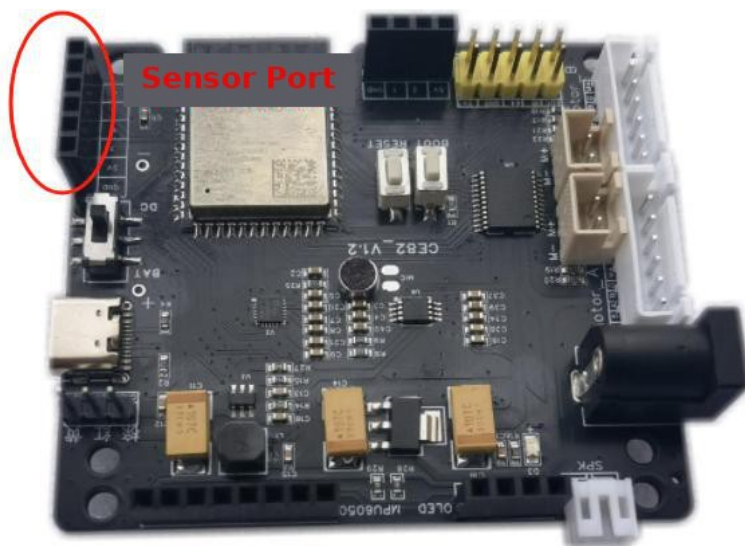
The ESP32 driver board with the display connected is shown below:





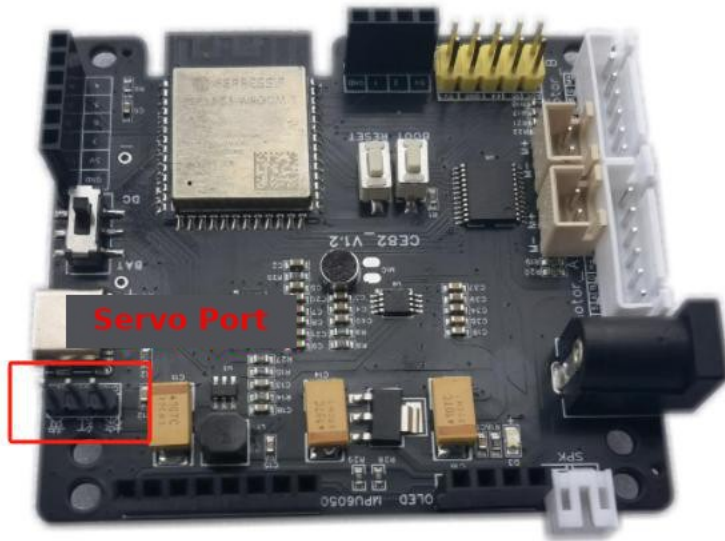
## 1.8. Sensor Port

This port is designed to connect a line-following module, a 400M RC receiver board, or a PS2 receiver board — only one of these modules can be connected at a time.



## 1.9. Servo Port

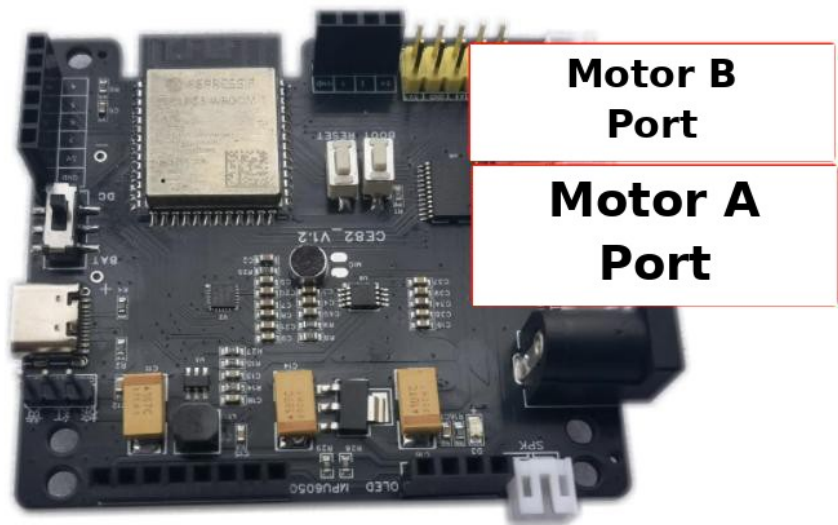
The board provides one servo channel. Match the servo wire colors to the brown/red/yellow labels on the board, as shown.



Note: on the servo port, yellow is the signal wire, red is 5V power, and brown is ground.

### 1.10. Motor Port

The board uses the ESP32 to control a TB6612 driver chip, which can drive 2 motor channels. Their positions are shown below:



Both Motor A and Motor B ports have two connectors: a 2-pin connector (for a plain motor) and a 6-pin connector (for an encoder motor), as shown:





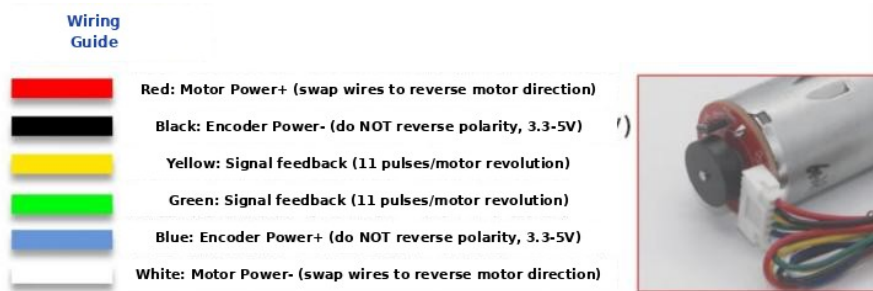
### Truth table for Motor A:

| ESP32 IO | 45 (IN1) | 48 (IN2) | 47                                                              | 14 / 21                                                                                       |
|----------|----------|----------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Forward  | Low      | High     | Outputs PWM for speed; duty=0 is slowest, higher duty = faster. | For encoder motors: reads the square-wave feedback from the encoder motor to calculate speed. |
| Reverse  | High     | Low      |                                                                 |                                                                                               |
| Stop     | Low      | Low      |                                                                 |                                                                                               |

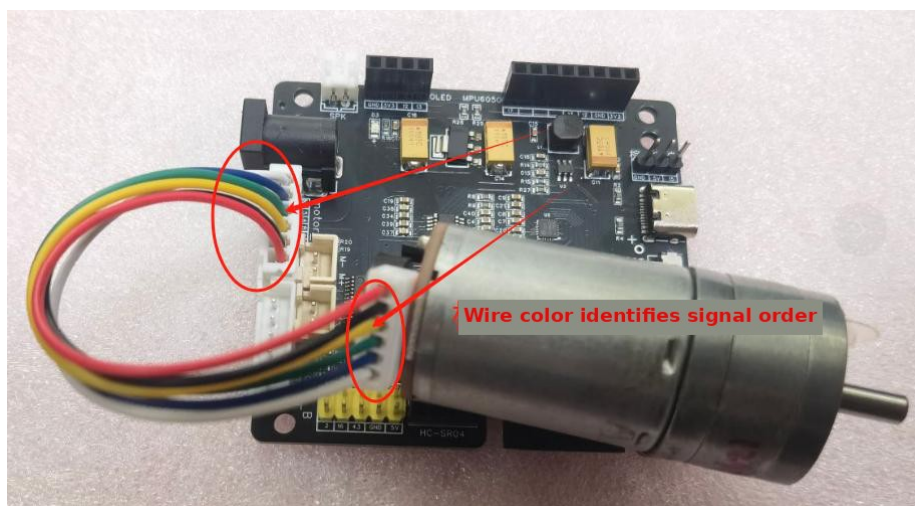
### Truth table for Motor B:

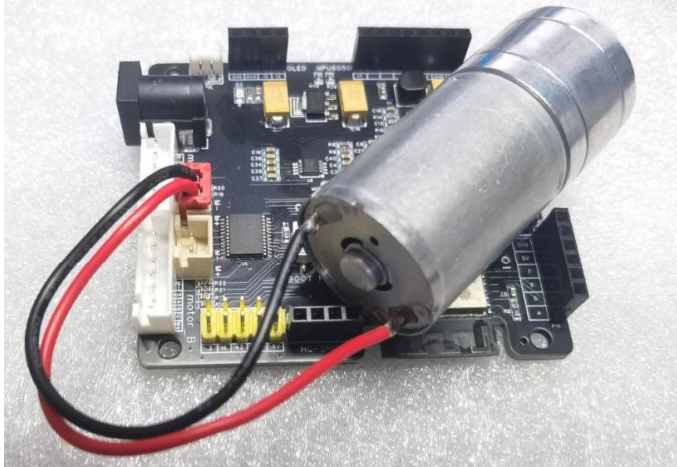
| ESP32 IO | 38 (IN1) | 39 (IN2) | 40                                                              | 41 / 42                                                                                       |
|----------|----------|----------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Forward  | High     | Low      | Outputs PWM for speed; duty=0 is slowest, higher duty = faster. | For encoder motors: reads the square-wave feedback from the encoder motor to calculate speed. |
| Reverse  | Low      | High     |                                                                 |                                                                                               |
| Stop     | Low      | Low      |                                                                 |                                                                                               |

Wire order for the 370 encoder motor is shown below:



Connect it to the board as shown below — either Port A or Port B works:



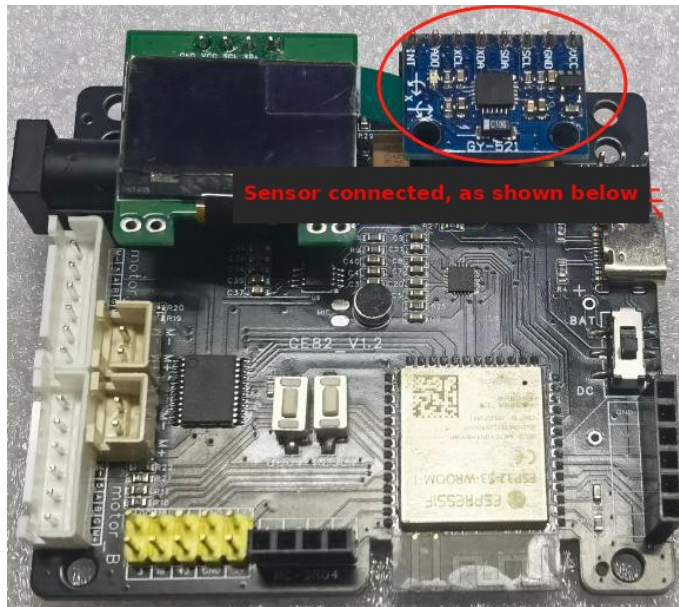


In the provided experiments, either Motor Port A or B can be used — whether you connect the 2-pin or 6-pin connector depends on the type of motor.

### 1.11. MPU6050 Port

The ESP32 driver board provides an MPU6050 6-axis sensor port. This sensor is used on two-wheel self-balancing cars. Connecting the sensor is shown below:

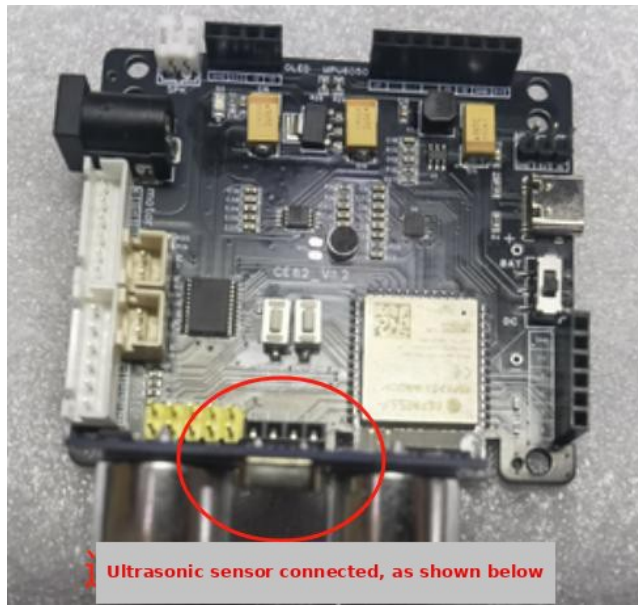




### 1.12. Ultrasonic Port

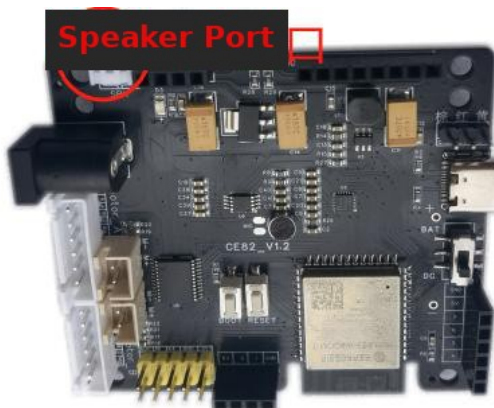
The ESP32 driver board provides an HC-SR04 ultrasonic sensor port. Connecting the sensor is shown below:





### 1.13. Speaker Port

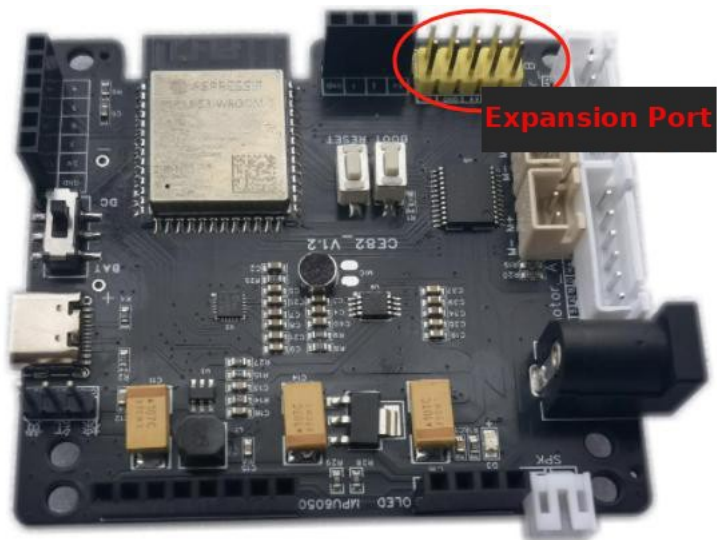
The ESP32 driver board provides a speaker port, enabling the AI voice-assistant feature — you can voice-control the car to move forward, reverse, change speed, turn, and stop. It uses an XH2.0 reverse-polarity-proof connector, as shown:



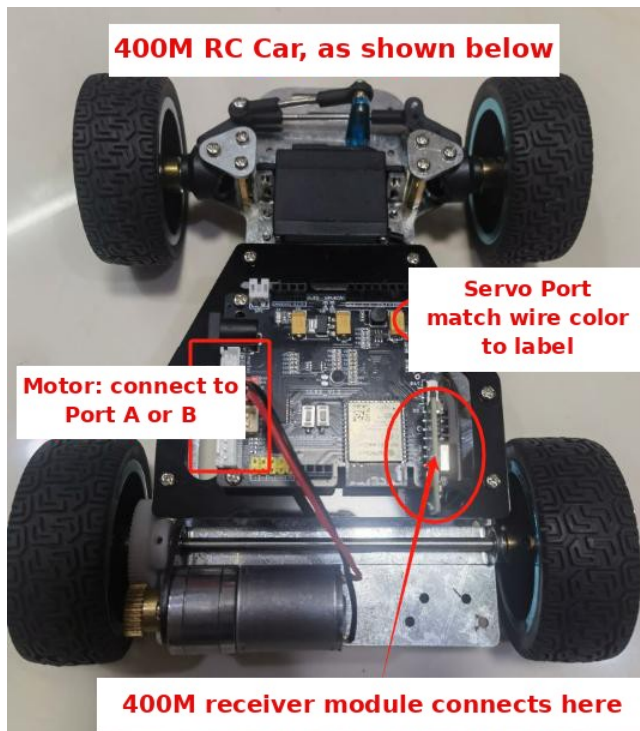
### 1.14. Expansion Port

The ESP32 driver board has several unused IO pins that are broken out to an expansion port, so users can connect their own custom sensors, etc. The unused IO pins are: 3, 16, 18, 43, and 44; each pin is silkscreened next to the expansion port. Note that the ADC pin and pin 3 are tied together, which lets you measure the battery voltage.

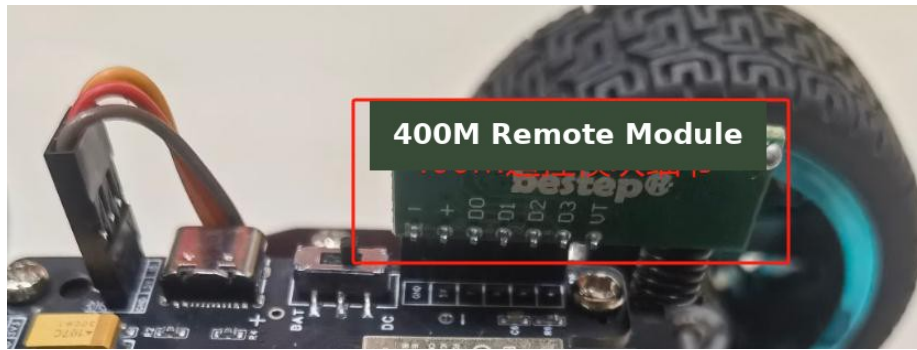




### 1.15. 400M RC Car — Fully Wired

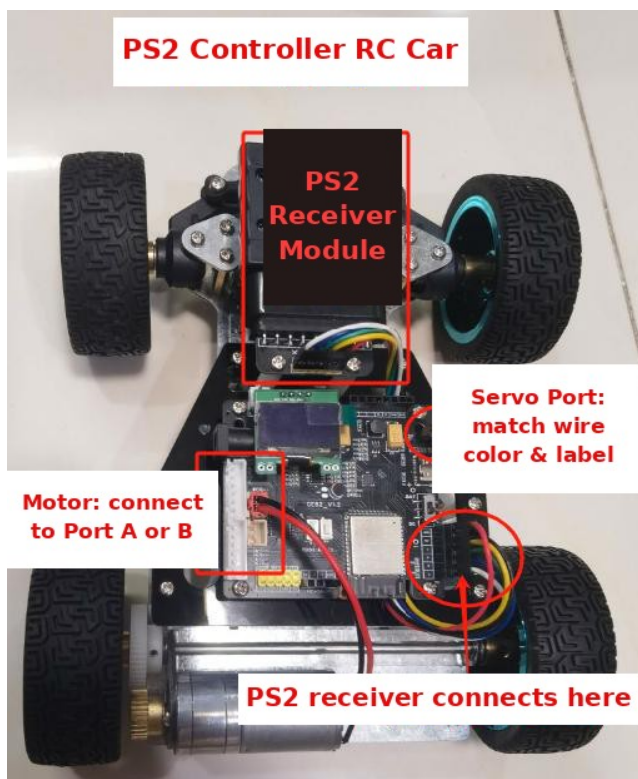






### 1.16. PS2 RC Car — Fully Wired

In the image below, the PS2 wireless receiver module wiring is: red (GND), black (5V), yellow (DAT – IO7), green (CMD – IO6), blue (CS – IO5), white (CLK – IO4).



### 1.17. Line-Following Car — Fully Wired

In the image below, the line-following module wiring is: red (GND), black (5V), yellow (S1 – IO7), green (S2 – IO6), blue (S3 – IO5), white (S4 – IO4).

**Line-following car, wired as shown**



**4-Channel Line  
Sensor Module**

**Servo Port:  
match wire color**

**Motor: connect  
to Port A or B**

